

Weight and printing

Paper-review result: cruise is plausible under the stated assumptions. Flight and hand launch remain on hold until stability/trim, propulsion and structural checks close. This is a screening calculation, not a flight clearance.

Installed category	Nominal mass
Printed parts	728 g
Plywood	243 g
Foam	337 g
Metal landing gear	144 g
Battery	512 g
Other equipment, tubes, wiring, glue and hardware	669 g
Total	2.63 kg

The estimate includes a 25 g propeller/nut allowance, 90 g glue/tape, 90 g fasteners/linkages, 80 g wiring, 70 g tubes/retainers and 75 g wheels. These are allowances to replace with actual weights. Plywood assumes 0.55 g/cm³; unpeeled foam assumes 400 g/m².

What is printed

Group	Installed plastic
Nose shell A59	150 g
Centre fairing A06 / tail housing A12	65 g / 122 g
Two ailerons / two hinge rails	115 g / 56 g
Two wing-root fittings / nose gear mount and bearing	56 g / 19 g
Mounts, spacers, washers, hatches and supports	144 g

There are 38 aircraft STL files for 42 installed pieces. The fit coupon is additional. Object extrusion totals 728 g; filament with automatic supports is 1.30 kg, or 1.50 kg with 15% waste allowance. Budget two 1 kg spools.

PrusaSlicer estimate: 0.4 mm nozzle, 0.2 mm layers, 4 top/bottom layers; shells 2 walls/5%, controls 3 walls/10%, carriers 4 walls/30%, automatic supports. A12 uses the shell setting only in this estimate; its structural process is not qualified. Reorient/reslice for the actual printer and reweigh. PLA+ is not the LW-PLA used in the video.

Balance, lift and stability

Wing area 0.30445 m²; span 2.000 m; mean aerodynamic chord 169.83 mm; aspect ratio 13.14. At 2.633 kg, weight is 25.83 N and wing loading is 86.5 g/dm².

Battery centre Y	Calculated CG	Fraction of MAC
-60 mm / forward stop	517.2 mm aft nose	22.0%
-33.7 mm	522.3 mm aft nose	25.0%
+20 mm / aft stop	532.8 mm aft nose	31.2%

CG = sum(mass x station) / total mass. The table is adjustment capability, **not an approved CG range**. Only about 36.6 g added at Y450 would consume the remaining ability to reach geometric quarter-MAC with the battery fully forward. Finish weighing before fixing the battery stops.

Assumed maximum lift coefficient	Stall speed	1.3 x stall reference
0.8	13.16 m/s	17.10 m/s
1.0	11.77 m/s	15.30 m/s
1.2	10.74 m/s	13.97 m/s

Stall speed = $\sqrt{2 \times \text{weight} / (\text{air density} \times \text{wing area} \times \text{maximum lift coefficient})}$. Density is 1.225 kg/m³ here. At 1.0 kg/m³, speeds increase by 10.7%. The lift coefficients are assumed; low-Reynolds-number airfoil behaviour, paper texture, twist and forming accuracy are not measured. Reynolds number is about 135,000 at 12 m/s and 230,000 at 20 m/s.

Tail and stability screen

The V-tail has about 0.0501 m² effective pitch area, 0.0246 m² effective yaw area, 35-degree cant and a 0.368 m tail arm. Pitch tail-volume coefficient is 0.357; vertical volume is 0.0149.

The linear model is SM = tail efficiency x lift-slope ratio x (1 - downwash gradient) x tail volume - body allowance - CG offset / MAC. Assumed factors are 0.8-1.0, 0.65-0.9, 0.25-0.5 and 0.03-0.10 MAC respectively.

Battery position	Static-margin sensitivity
Forward stop	+2.3% to +24.1% MAC
Geometric quarter-MAC	-0.7% to +21.1% MAC
Aft stop	-6.9% to +15.0% MAC

These are extreme assumption cases, not measured stability limits. They neither establish a safe aft limit nor prove the actual aircraft unstable. Resolve neutral point, trim, incidence and combined V-tail authority before flight.

Power, launch and flight time

Selected hardware remains Emax ECO II 2807 1300KV, Gemfan 9045CL three-blade, 4S 5200 mAh and a 40 A ESC. No verified thrust/current map for this exact combination is available in this package. The selection is a build assumption; it has not passed a propulsion test.

Nominal cruise scenario	Electrical power	Current at 14 V
16 m/s	61-147 W	4.4-10.5 A
20 m/s	98-257 W	7.0-18.3 A

$q = 0.5 \times \text{density} \times \text{speed squared}$; $CL = \text{weight} / (q \times \text{area})$; $CD = CD0 + CL \text{ squared} / (\pi \times \text{efficiency factor} \times \text{aspect ratio})$; $\text{drag} = q \times \text{area} \times CD$; $\text{electrical power} = \text{drag} \times \text{speed} / \text{propulsion efficiency} + 10 \text{ W avionics}$.

Assumptions: zero-lift drag coefficient 0.035-0.070, span efficiency 0.65-0.85 and combined propulsion efficiency 0.45-0.65. These broad scenarios are not a current, climb-rate or thrust prediction. Loaded motor RPM and propeller thrust at flying speed remain unknown.

Launch airspeed	Lift coefficient needed for 1g flight
8 m/s	2.16
10 m/s	1.39
12 m/s	0.96
14 m/s	0.71

An 8-10 m/s throw is not sufficient for steady 1g flight if maximum lift is within the assumed 0.8-1.2 range. Acceleration and height loss depend on thrust versus airspeed, attitude, wind and technique. An autopilot launch trigger is not flying speed. No ordinary hand launch is cleared by this calculation.

Flight-time planning

Average total battery current	Time with 20% nominal charge reserve
8 A / 12 A	31.2 min / 20.8 min
16 A / 20 A	15.6 min / 12.5 min
25 A / 40 A	10.0 min / 6.2 min

Planning range: about 12-20 minutes only if average total current is 12-20 A. $\text{Time} = 5.2 \text{ Ah} \times 0.8 \times 60 / \text{current} = 249.6 / \text{current}$, in minutes. Nominal battery energy is 76.96 Wh; 80% is 61.57 Wh. Capacity loss, climb, wind and return reserve affect the result. Use calibrated flight logs to select a real endurance estimate.

Controls and structural loads

MG90S manufacturer stall torque at 4.8 V is 1.8 kgf-cm, or 0.1765 Nm. The screening allowance below is 50% of stall: 0.0883 Nm. This is an assumption, not a continuous servo rating.

Control at 20 m/s	Hinge torque	Servo torque
Aileron: 0.01787 m ² , 74.3 mm chord	0.0651 Nm	0.0434 Nm
Moving tail: 0.01060 m ² , 40 mm chord	0.0208 Nm	0.0139 Nm

Hinge moment = $q \times \text{control area} \times \text{control chord} \times \text{hinge coefficient}$. Assumed hinge coefficient is 0.2; servo arm is 10 mm and control horn 15 mm. Servo torque = hinge moment $\times 10/15$. Aileron demand rises to 0.0678 Nm at 25 m/s and 0.0976 Nm at 30 m/s, above the screening allowance. No speed or flutter limit is established.

An ideal straight linkage gives about 28 degrees of control rotation at 45 degrees servo rotation, or 35 degrees at 60 degrees. Real horn alignment, friction and combined V-tail mixing change this. Verify simultaneous pitch and yaw commands, mechanical stops and voltage at the farthest servo. Vx is rated 5 A continuous/6 A peak; the separate peripheral 5 V rail is 2 A.

Wing and joiners

Screening case	Demand
Uniform 1g lift / each half-wing root	6.46 Nm
Uniform 4g lift / each half-wing root	25.83 Nm
Elliptical 1g / 4g lift	5.48 / 21.92 Nm
Two 8/6 mm tubes sharing 4g moment equally	About 376 MPa per tube
One tube carrying that entire moment	About 752 MPa

Uniform root moment = total weight $\times$ half-span / 4. A single 8 mm OD / 6 mm ID tube has elastic section modulus 34.36 mm³; stress = moment / section modulus. These are load illustrations, not the actual load split or tube strength. Spars, sleeves, bonds, printed roots and skins also carry load. Tube layup, joint stiffness and crush strength require evidence; 5 mm solid rods are not substitutes.

A12 is a structural tail housing

The fixed tail meets A12 and is about 3 mm from plywood bulkhead A18. Tail loads therefore need a verified transfer through the printed slot walls and bonded housing into the frame. A12 also supports two servo mounts. Do not treat its 2-wall/5% slicing estimate as a qualified production recipe. Check local wall orientation, root bonds, fasteners and the A12-to-A18 joint using representative coupons and the assembled tail.

Landing loads and closure checks

Current 76.2 mm wheel gear	9-inch propeller clearance
Rigid static position	143 mm
10 degrees nose-up from static	59 mm
10 degrees, less 15 mm assumed deflection and 25 mm grass	19 mm
12 degrees, with those same deductions	About 2 mm

These are geometry cases, not measured deflection or a landing rating. At nominal balance the static distribution is approximately 13% nose wheel and 44% on each main wheel. Pitch, grass, rebound, side loads and uneven ground can consume the clearance.

For the 20 x 4 mm 6061-T6 main strip, elementary 3g bending gives about 90 MPa at 2.63 kg. Stress = wheel force x lever arm / section modulus; the lever arm is about 142 mm and strip section modulus 53.3 mm³. Specified stock yield is at least 240 MPa. This straight-beam comparison does not qualify bends, bolts, wood bearing or impact.

Assumed sink / stopping stroke	Vertical energy	Mean total reaction
1 m/s / 20 mm	1.32 J	About 3.6g
2 m/s / 20 mm	5.27 J	About 11.2g
2 m/s / 10 mm	5.27 J	About 21.4g

Vertical energy = 0.5 x mass x sink speed squared. Mean reaction = energy / stroke + weight. These scenarios explain why a 3g strip calculation is not a hard-landing proof. Horizontal flight energy is largely dissipated through aerodynamic braking and rollout, not the same vertical gear stroke.

Before flight	Evidence to record
Mass and CG	Finished component weights, complete-aircraft balance, secured battery position and a justified flight CG range.
Controls and structure	Full combined control travel; loaded servo supply; joined wing/root and A12 tail bending/torsion checks; no residual damage.
Propulsion and launch	Actual current, thrust, voltage sag and temperature with selected propeller; demonstrated launch method and pilot recovery plan.
Systems and landing	Sensor calibration, radio range/failsafes, loaded compass interference, gear stroke, clearance and fastener retention.

Calculation sources

NASA lift/drag | NASA induced drag | TowerPro MG90S | Matek F405-WING V2. Geometry is from the supplied CAD. Material properties, aerodynamic coefficients and unweighed hardware are explicitly assumed. No ground or flight test is represented as completed.